Wentao Zhang

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- Date of Birth: Nov. 1995 · Address: Suzhou, China

Work Experience

University of Nottingham 2025/09 - Now

Research Assistant Nottingham, United Kingdom

- Power Electronics and Machines Centre (PEMC)

Education

Southeast University 2021/09 - 2025/09

Ph. D. in Electrical Engineering Nanjing, Jiangsu, China

- Dissertation Title: "Multi-physics Analysis and Design of Hairpin Winding PM Traction Machines in Electric Vehicles"
- 2025 1st Prize, IEEE TEC Electric Vehicle Design Competition
- 2025 2nd Prize Paper Award, IEEE Transactions on Transportation Electrification

Southeast University 2018/09 - 2021/06

Master in Electrical Engineering Nanjing, Jiangsu, China

- Dissertation Title: "Analysis and Design of High-Performance Wound-Excited Magnetic Field Modulation-based Brushless Machines"

Southeast University 2014/09 - 2018/06

B.Eng in Electrical Engineering Nanjing, Jiangsu, China

- Dissertation Title: "Comparative Study of Control Systems for PM Machine in Electric Vehicles"

Research Experience

My research focuses on AI-driven multiphysics design, digital-twin modelling and advanced manufacturing of high power-density electrical machines for electrified transport, combining strongly coupled electromagnetic, thermal, fluid and mechanical finite-element analysis with deep-learning surrogate models and experimental validation on dynamometer-tested prototypes. I have authored 22 journal papers, including seventeen published IEEE Transactions papers and one more in early access, 18 peer-reviewed conference papers, and 9 patents. The work groups into four themes.

Theme 1, novel manufacturing-ready machine architectures, covers the n-layer flat-wire and hairpin winding design methodology for high power-density EV PM machines, conductor-layout optimisation, mechanical and tolerance-aware design, and full prototype validation. **Theme 2, multi-physics design and coupled cooling**, covers high-resolution lumped-parameter thermal networks and fluid-solid coupled oil-cooling models for hairpin-winding PM machines, validated against AVL dynamometer measurements. **Theme 3, AI-driven surrogate modelling and digital twins**, covers an implicit-physics deep-learning surrogate that maps machine design parameters to the magnetic vector potential field, reaching $R^2 = 0.9996$ for average torque and 0.9730 for iron loss while reducing inference time by more than 94% against direct finite-element analysis, built on an in-house Python finite-element code package covering electromagnetic, structural-modal and mechanical-stress analyses. **Theme 4, machine control and dynamic modelling**, covers spatial-harmonic inductance modelling, DC winding current dynamics under PWM excitation, and converter-machine interaction analysis for wound-field and PM machine systems.

Industrial development experience. In partnership with Inovance, a world-leading EV traction-machine supplier, I have analysed, designed, manufactured and AVL-dynamometer-tested two advanced oil-cooled PM traction machines: an 18 000 rpm 113 kW A0-class machine (138 Nm, 7.78 kW/kg, 95.3% peak efficiency) and a 20 000 rpm 270 kW A-class machine (380 Nm, 7.87 kW/kg, 97.4% peak efficiency). Each was taken through a full industrial cycle of multi-physics design, engineering drawings and tolerance specification, and dynamometer characterisation.

Software and toolchain. In-house Python finite-element code package (electromagnetic, structural-modal, mechanical-stress) used as the substrate for the deep-learning surrogate above; deep-learning surrogate model in PyTorch; MATLAB/Simulink; ANSYS Maxwell and Mechanical; C/C++ for TI C2000 and STM32.

Research Publication

Transaction/Journal Paper

[J1] **Wentao Zhang**, W. Hua, Z. Wu, G. Zhao, Y. Wang, and W. Xia, "Analysis of DC Winding Induced Voltage in Wound-Field Flux-Switching Machine With Air-Gap Field Modulation Principle," *IEEE Trans. Ind. Electron.*, vol. 69, no. 3, pp. 2300–2311, Mar. 2022.

[J2] **Wentao Zhang**, Z. Wu, W. Hua, Z. Zhu, Z. Wu, J. Hu, et al., "Reduction of Open-Circuit DC Winding Induced Voltage and Torque Pulsation in the Wound Field Switched Flux Machine by Stator Axial Pairing of Tooth Tips," *IEEE Trans. Ind. Appl.*, vol. 58, no. 2, pp. 1976–1990, Mar.–Apr. 2022.

[J3] **Wentao Zhang**, Z. Wu, Y. Fan, W. Hua, and M. Cheng, "A Stable and Computationally Efficient Spatial Harmonic Model for Predicting the DC Winding Induced Voltage in WFSF Machine," *IEEE Trans. Ind. Appl.*, vol. 59, no. 4, pp. 3966–3977, Jul.–Aug. 2023.

[J4] **Wentao Zhang**, Z. Wu, Y. Fan, W. Hua, and M. Cheng, "Analysis and Improvement of DC Winding Current Dynamics in Wound Field Switched Flux Machine," *IEEE Trans. Ind. Appl.*, vol. 60, no. 4, pp. 6023–6032, Jul.–Aug. 2024.

[J5] **Wentao Zhang**, Z. Wu and W. Hua, "**Design of n-Layer Flat Wire Winding in Permanent Magnet Machine for Electric Vehicles**," *IEEE Trans. Transport. Electrific.*, vol. 11, no. 3, pp. 7798–7810, 2025.

[J6] **Wentao Zhang**, Z. Wu, L. Jin, Y. Fan, W. Hua, and M. Cheng, "Analysis and Multi-Objective Optimization of the Hybrid Excitation Switched Flux Machine," *IEEE Trans. Ind. Appl.*, vol. 61, no. 2, pp. 2996–3006, Mar.–Apr. 2025.

[J7] **Wentao Zhang**, Z. Wu, Z. Wang, W. Hua, Y. Fan, Z. Xu, et al., "Design Tradeoff Between Flux Regulation Capability and DC Winding Induced Voltage in Hybrid Excitation Switched Flux Machine," *IEEE Trans. Ind. Appl.*, vol. 61, no. 5, pp. 7198–7208, Sep. 2025.

[J8] **Wentao Zhang**, Y. Fan, Z. Zhu, Z. Wu, W. Hua, and M. Cheng, "Analysis of DC Winding Induced Voltage in Wound-Rotor Synchronous Machines by Using the Air-Gap Field Modulation Principle," *World Electr. Veh. J.*, vol. 13, no. 11, Art. no. 215, 2022.

[J9] **Wentao Zhang**, Z. Wu, Z. Cang, W. Hua, Y. Fan, and M. Cheng, "Influence of End-Effect in Switched Flux Machines," *IEEE Trans. Ind. Appl.*, vol. 62, no. 1, pp. 785–798, Jan. 2026.

[J10] **Wentao Zhang**, Z. Wu and W. Hua, "**High-Fidelity Lumped Parameter Thermal Model for Oil-Cooling Hairpin Winding PM Machines in EVs**," *IEEE Trans. Transport. Electrific.*, vol. 12, no. 2, pp. 2978–2991, Apr. 2026.

[J11] Z. Wu, **Wentao Zhang**, Y. Fan, W. Hua, and M. Cheng, "A Transient-State Lumped Parameter Thermal Model for Brushless Wound Field Switched Flux Machines," *IEEE Trans. Transport. Electrific.*, vol. 10, no. 1, pp. 224–238, Mar. 2024.

[J12] Z. Wu, Z. Zhu, C. Wang, W. Hua, K. Wang, and **Wentao Zhang**, "Influence of Rotor Iron Bridge Position on DC-Winding-Induced Voltage in Wound Field Switched Flux Machine Having Partitioned Stators," *Chin. J. Electr. Eng.*, vol. 7, no. 3, pp. 20–28, Sep. 2021.

[J13] Z. Wu, Z. Zhu, S. Cai, W. Hua, and **Wentao Zhang**, "Enhancement of Torque Density in Wound Field Switched Flux Machines With Partitioned Stators Using Assisted Ferrites," *Chin. J. Electr. Eng.*, vol. 7, no. 3, pp. 42–51, Sep. 2021.

[J14] Z. Wu, L. Jin, W. Hua, **Wentao Zhang**, and M. Cheng, "Harmonics Orders Modeling of DC Winding Induced Voltage Pulsation in Wound Field Switched Flux Machines Under PWM Excitation," *IEEE Trans. Transport. Electrific.*, vol. 10, no. 3, pp. 7040–7050, Sep. 2024.

[J15] Z. Wu, L. Jin, **Wentao Zhang**, Y. Fan, W. Hua, and M. Cheng, "Influence of PWM Excitation on DC Winding Induced Voltage Pulsation in Wound Field Switched Flux Machines," *IEEE Trans. Ind. Appl.*, vol. 60, no. 1, pp. 460–476, Jan.–Feb. 2024.

[J16] Z. Wu, Z. Zhu, W. Hua, S. Akehurst, X. Zhu, **Wentao Zhang**, et al., "Analysis and Suppression of Induced Voltage Pulsation in DC Winding of Five-Phase Wound-Field Switched Flux Machines," *IEEE Trans. Energy Convers.*, vol. 34, no. 4, pp. 1890–1905, Dec. 2019.

[J17] L. Jin, W. Hua, U. B. Akuru, Z. Wu, **Wentao Zhang**, and M. Cheng, "Comparative Study on DC Winding Induced Voltage Pulsation of Wound Field Flux Modulation Machines Having Different Iron Core Structures," *IEEE Trans. Ind. Appl.*, vol. 61, no. 2, pp. 2874–2883, Mar.–Apr. 2025.

[J18] X. Yan, Z. Wu, W. Hua, **Wentao Zhang**, and M. Cheng, "A Mathematical Model for Wound Field Switched Flux Machine Considering Inductance Harmonics of Field and Armature Windings," *IEEE Trans. Ind. Appl.*, vol. 60, no. 5, pp. 6811–6822, Sep.–Oct. 2024.

[J19] H. Hua, X. Chen, D. Li, **Wentao Zhang**, Z. Wu, and W. Hua, "Investigation of Torque Improvement of Modular Stator Surface-Mounted Permanent Magnet Machines With Flux Gaps," *IEEE Trans. Ind. Appl.*, vol. 60, no. 5, pp. 6787–6798, Sep.–Oct. 2024.

[J20] J. Chen, W. Hua, L. Shao, Z. Wu, X. Yan, and **Wentao Zhang**, "Modified Magnetic Equivalent Circuit of Double-Stator Single-Rotor Axial Flux Permanent Magnet Machine Considering Stator Radial-End Flux-Leakage," *IET Electr. Power Appl.*, vol. 18, no. 2, pp. 195–207, Feb. 2024.

[J21] Z. Wu, X. Yan, Z. Wang, **Wentao Zhang**, and W. Hua, "**Influence of PWM Excitation on Loss and Temperature of Flat Wire Permanent Magnet Machines for Electric Vehicles**," *IEEE Trans. Ind. Electron.*, vol. 73, no. 4, pp. 5174–5185, Apr. 2026.

[J22] Z. Wu, E. Fu, J. Du, **Wentao Zhang**, and W. Hua, "Quantitative Comparison of Electromagnetic Vibration of PMSMs With Glued and Welded Stator Core for Electric Vehicles," *IEEE Trans. Ind. Appl.*, early access, 2025.

Conference Paper

[C1] **Wentao Zhang**, Zhongze Wu and Wei Hua, "**A Deep Learning Based Magnetic Field Surrogate Modeling Method for Interior Permanent Magnet Machine in EVs**," in Proc. Int. Conf. Elect. Mach. (ICEM), Funchal, Madeira, Portugal, 2026. (**Accepted**)

[C2] **Wentao Zhang**, Zhongze Wu, Wei Hua and Ziqiang Zhu, "Reduction of Open-Circuit DC Winding Induced Voltage and Torque Pulsation in the Wound Field Switched Flux Machine by Stator Axial Pairing of Tooth-Tips," in Proc. IEEE Int. Elect. Mach. Drives Conf., Gothenburg, Sweden, 2020, pp. 522-528.

[C3] **Wentao Zhang**, Zhongze Wu, Ying Fan, Wei Hua and Ming Cheng, "A Stable and Computationally Efficient Spatial Harmonic Model for Predicting the DC Winding Induced Voltage in WFSF Machine," in Proc. IEEE Energy Convers. Congr. Expo., Detroit, MI, USA, 2022, pp. 1-7.

[C4] **Wentao Zhang**, Zhongze Wu, Wei Hua, Ying Fan, Zhiyuan Xu and Ming Cheng, "Design Tradeoff between Flux Regulation Capability and DC Winding Induced Voltage in Hybrid Excitation Switched Flux Machine," in Proc. IEEE Energy Convers. Congr. Expo., Nashville, TN, USA, 2023, pp. 5060-5066.

[C5] **Wentao Zhang**, Zhongze Wu, Lai Jin, Ying Fan, Wei Hua and Ming Cheng, "Analysis and Multi-Objective Optimization of the Hybrid Excitation Switched Flux Machine," in Proc. Int. Elect. Energy Conf., Hefei, China, 2023, pp. 3371-3376. (**Best Paper Award**)

[C6] **Wentao Zhang**, Zhongze Wu, Wei Hua, Ying Fan and Ming Cheng, "Influence of the End-Effect in Wound Field Switched Flux Machine," in Proc. Int. Conf. Elect. Mach. Syst., Zhuhai, China, 2023, pp. 1223-1227.

[C7] **Wentao Zhang**, Zhongze Wu, Ying Fan, Wei Hua and Ming Cheng, "Analysis and Improvement of DC Winding Current Dynamics in Wound Field Switched Flux Machine," in Proc. Int. Conf. Energy Technol. Future Grids, Wollongong, Australia, 2023, pp. 1-6.

[C8] **Wentao Zhang**, Zhongze Wu, Wei Hua and Ji Qi, "Analysis of PM Volume Reduction Methods for Hybrid Excitation Switched Flux Machines," in Proc. Int. Conf. Elect. Mach. Syst., Zhuhai, China, 2024, pp. 3152-3157.

[C9] **Wentao Zhang**, Zhongze Wu and Wei Hua, "Field-Oriented Design of Notching and Mechanical Stress Relieving Grooves for Rotors in Flat Wire Permanent Magnet Machines," in Proc. IEEE Int. Elect. Mach. Drives Conf., 2025, pp. 304-309.

[C10] Zhongze Wu, Lai Jin, **Wentao Zhang**, Ying Fan, Wei Hua and Ming Cheng, "Influence of PWM Excitation on DC Winding Induced Voltage in Wound Field Switched Flux Machines," in Proc. IEEE Energy Convers. Congr. Expo., Detroit, MI, USA, 2022, pp. 1-7.

[C11] Lai Jin, Wei Hua, Udochukwu B. Akuru, Zhongze Wu, **Wentao Zhang** and Ming Cheng, "Comparative Study on DC Winding Induced Voltage Pulsation of Wound Field Flux Modulation Machines Having Different Iron Core Structures," in Proc. Int. Conf. Elect. Mach. Syst., Zhuhai, China, 2023, pp. 442-447.

[C12] Martin Philip Koroma, Wei Hua, Zhongze Wu, Lai Jin and **Wentao Zhang**, "Influence of Magnet Layer Numbers on Electromagnetic Performance of Interior Permanent Magnet Machines," in Proc. Int. Conf. Elect. Mach. Syst., Zhuhai, China, 2023, pp. 696-700.

[C13] Xinying Chen, Hao Hua, Dalin Li, **Wentao Zhang**, Zhongze Wu and Wei Hua, "Modular Stator Surface-Mounted Permanent Magnet Machines with Flux Gaps," in Proc. IEEE Int. Elect. Mach. Drives Conf., San Francisco, CA, USA, 2023, pp. 1-7.

[C14] Xueyi Yan, Zhongze Wu, Wei Hua, **Wentao Zhang** and Ming Cheng, "A Mathematical Model for Wound Field Switched Flux Machine Considering Inductance Harmonics of Field and Armature Windings," in Proc. IEEE Int. Elect. Mach. Drives Conf., San Francisco, CA, USA, 2023, pp. 1-6.

[C15] Zhongze Wu, Lai Jin, **Wentao Zhang**, Xueyi Yan, Haorui Ge, Hao Hua, Wei Hua and Ming Cheng, "A Lumped Parameter Thermal Model for Variable Flux Reluctance Machine Considering Relative Position of DC and AC Windings," in Proc. Int. Conf. Elect. Mach., Torino, Italy, 2024, pp. 1-7.

[C16] Zhongze Wu, Xueyi Yan, Zhimian Wang, **Wentao Zhang** and Wei Hua, "Quantitative Analysis of the Torque-Speed Curve for a Flat Wire Permanent Magnet Machine Drive System," in Proc. IEEE Int. Elect. Mach. Drives Conf., 2025, pp. 40-44.

[C17] Zhongze Wu, Erfang Fu, Jinwen Du, **Wentao Zhang** and Wei Hua, "Influence of Stator Core on the Vibration Behavior of Permanent Magnet Machines with Hairpin Winding," in Proc. IEEE Int. Elect. Mach. Drives Conf., 2025, pp. 34-39.

[C18] Zhongze Wu, **Wentao Zhang**, Zhimian Wang, Lingyun Shao, Wei Hua and Ming Cheng, "Stator Semi-Skewing Method for High-Speed Permanent Magnet Machine," in Proc. Int. Conf. Elect. Mach. Syst., 2025, pp. 693-697.

Patent

- Zhongze Wu,**Wentao Zhang**, Lai Jin, and Wei Hua, "Inductive Ripple Voltage Suppression Method Based on Low-Pass Filter Circuit," CN118508832A, 2024.
- Zhongze Wu, **Wentao Zhang**, Lai Jin, and Wei Hua, "Disk Brushless Electric Excitation Magnetic Field Modulation Motor With Double-Row Stator Structure," CN118100572A, 2024.
- Zhongze Wu, **Wentao Zhang**, Huilin Huang, and Wei Hua, "Method for Constructing a High-Precision Motor Performance Database Based on a Dual-Model Strategy," CN117972924A, 2024.
- Zhongze Wu, **Wentao Zhang**, Wei Hua, and Zhixiang Zou, "Method for Establishing General Lumped Parameter Thermal Network Model of Motor," CN117094088A, 2023.
- Zhongze Wu, **Wentao Zhang**, Lai Jin, and Wei Hua, "Method for Analyzing Harmonic Orders of Induced Ripple Voltage of Excitation Winding," CN115642847A, 2023.
- Zhongze Wu, **Wentao Zhang**, Junxiang Liao, Ying Fan, Wei Hua, and Ming Cheng, "Excitation Winding Induced Pulse Voltage Suppression Method Based on Harmonic Current Injection," CN115208245A, 2022.
- Zhongze Wu, **Wentao Zhang**, Ying Fan, Wei Hua, and Ming Cheng, "High-Precision Field Path Coupling Design Method for Brushless Electric Excitation Magnetic Field Modulation Motor," CN114781227A, 2022.
- Zhongze Wu, **Wentao Zhang**, Wei Hua, and Ziqiang Zhu, "Method for Weakening the Induced Pulsating Voltage of a Brushless Electric Excitation Magnetic Field Modulated Motor," CN114189185A, 2022.
- Wei Hua, **Wentao Zhang**, Zhongze Wu, and Shichuan Ding, "Method for Reducing Torque Ripple and Induced Voltage of Electric Excitation Flux Switching Motor," CN111769697A, 2020.

Academic Service

- **Co-Organiser and Co-Chair**, ICEM 2026 Special Session on "Next-Generation Electrified Transport: Advances in Electric Drive Systems", with Dr. Lingyun Shao (NUAA), Funchal, Madeira, Portugal, Sep. 2026.
- Peer reviewer for IEEE journals, including *IEEE Trans. Industrial Electronics*, *Industry Applications*, *Energy Conversion*, and *Transportation Electrification*.

Awards

- 2025 1st Prize, IEEE TEC Electric Vehicle Design Competition
- 2025 2nd Prize Paper Award, IEEE Transactions on Transportation Electrification
- 2024 2nd Prize in 11th Energy Equipment Innovative Design Competition for China Postgraduate
- 2023 **Wiscom System Scholarship**
- 2023 1st Prize Academic Scholarship
- 2023 CIEEC Best Paper Award
- 2023 **1st Prize in IEEE IAS CMD Thesis Contest (non-Ph. D)**
- 2022 Outstanding Master Degree Thesis in Jiangsu Province
- 2017 2nd Prize in Southeast University Smart Car Competition

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